PRD & SPECIFICATION

Personal Health & Progress Dashboard

Document Version: 1.0 (MVP)
Platform: Google Sheets / Excel

PRIMARY METRIC	INPUT FREQUENCY	DESIGN AESTHETIC	ANALYTICS ENGINE
Weight & Trends	Daily (1 Sheet)	Blush / Cream	Automated

1. Product Overview & Vision

The **Personal Health & Progress Dashboard** is a reusable, formula-driven Google Sheets tracking tool designed to record daily health metrics, monitor progress toward target goals, and surface actionable insights through smoothed 7-day trendlines and visual analytics. Users enter data on a single daily log, while all dashboard metrics, milestones, and charts update dynamically.

2. CORE OBJECTIVES

- **Single-Point Data Entry:** Log daily weight, calories, steps, workouts, water, sleep, and mood without navigating multiple tabs.
- **Noise Reduction via Moving Averages:** Smooth daily body weight fluctuations using an automated 7-day rolling average.
- **Goal & Progress Tracking:** Calculate percentage completed, remaining weight change, and target date countdowns.
- **Zero Formula Maintenance:** Allow end-users to operate the dashboard without editing or writing complex logic.
- **Dual Demo/Template Architecture:** Provide pre-filled realistic 30-day example data alongside a clean blank sheet.

3. WORKBOOK ARCHITECTURE & STRUCTURE

Sheet Identifier	Sheet Name	Functional Purpose
01 — Dashboard	Main KPI Overview	Executive summary screen with KPI cards, goal trackers, milestones, and embedded trend charts.
02 — Daily Log	Primary Input Log	Central structured data entry table with pre-built 7-day rolling average calculation formulas.
03 — Goals & Settings	Configuration Area	Centralized configuration tab for starting metrics, calorie targets, step goals, and target dates.
04 — Analytics	Statistical Engine	Deep analytical summary containing period comparisons, adherence rates, and min/max stats.
05 — Example Data	Demonstration Suite	Pre-populated 30-day realistic sample dataset demonstrating complete template functionality.
06 — Instructions	User Guide	Clear, 5-step beginner instructions on copying, configuring, and operating the workbook.

4. FUNCTIONAL REQUIREMENTS (FR-01 TO FR-10)

Req ID	Feature Module	Functional Requirement Specification
FR-01	Daily Data Entry	Provide one structured sheet for date, weight, calories, steps, workout, water, sleep, mood, and notes.
FR-02	Weight Tracking	Capture daily weight; dynamically calculate starting weight, current weight, total change, and min/max.
FR-03	Calorie Analytics	Log intake; compute daily/weekly averages, target variance, and adherence percentage within target range.
FR-04	Activity Tracking	Track step count and binary workout status; compute workout consistency and weekly average steps.
FR-05	Water & Sleep	Log daily water (ml) and sleep (hours); aggregate weekly averages and compare against user targets.
FR-06	Mood Selection	Provide pre-defined emoji dropdown list (😍 Great, 😊 Good, 😐 Okay, 😕 Low, 😩 Bad) for simple tracking.
FR-07	Goals & Targets	Allow user configuration of Start Weight, Target Weight, Daily Calorie Goal, and Step Targets.
FR-08	Goal Progress %	Compute distance between start weight and current weight vs. total distance to target weight.
FR-09	Countdown Engine	Calculate target date, days remaining, elapsed days, and percentage of target time elapsed.
FR-10	Automated Charts	Render a dual-line Weight Trend Chart (Daily vs. 7-Day Smoothed) and Calorie Bar Chart.

5. MATHEMATICAL LOGIC & ANALYTICS ENGINE

7-Day Rolling Moving Average Calculation:

```
7-Day Weight Avg = AVERAGE(Daily_Weight[Day_0] : Daily_Weight[Day_-6])
```

To prevent daily water weight noise from creating false feedback, the dashboard plots the 7-day moving average as the primary trend indicator. Target adherence percentage is calculated as:

```
Adherence % = (Count of Days within ±100 kcal of Goal / Total Logged Days) * 100
```

6. VISUAL DESIGN SYSTEM & PALETTE (UI-01 TO UI-04)

Color Name	Hex Code	Role	Application Area
Blush Pink	#E88D93	Primary Accent	Sheet headers, main KPI highlights, primary buttons.
Warm Cream	#FAF6F0	Page Background	Workbook canvas, sheet background, card backgrounds.
Butter Yellow	#FFF2CC	Highlight / Status	Milestone cards, achievement badges, active states.
Muted Warm Brown	#4A3B32	Primary Typography	All table body text, metric labels, data values.

7. MILESTONE CALCULATION ENGINE (FR-19)

8. QUALITY ASSURANCE & EDGE CASE SPECIFICATIONS (SECTION 15)

Edge Case Scenario	Potential Risk	System Handling & Mitigation Strategy
Missing Days in Log	Formula breakdown (`#VALUE!`)	Charts skip blank entries without treating them as zero; rolling averages ignore empty rows.
Zero / Negative Inputs	Corrupted statistics	Data validation rules reject negative calories, negative weights, and invalid step counts.
Duplicate Dates Logged	Distorted weekly averages	Conditional formatting highlights duplicate date rows in bright yellow for user correction.
Empty Optional Fields	Broken KPI display	KPI cards use `IFERROR` wrappers to display default placeholder dashes (`--`) instead of errors.
Start Weight = Target Weight	Division by zero (`#DIV/0!`)	Goal progress formula validates that start and target weight are not equal prior to division.
Single Day Entry Only	Chart rendering errors	Rolling average formula gracefully outputs daily weight until 7 days of historical logs accumulate.

9. DATA VALIDATION & INTEGRITY RULES (DV-01 TO DV-04)

- **Weight Entry Validation:** Numeric value constrained between 30.0 kg and 300.0 kg.
- **Calorie Entry Validation:** Integer value constrained between 500 kcal and 10,000 kcal.
- **Step Count Validation:** Integer value constrained between 0 and 100,000 steps.
- **Dropdown Constraints:** Workout status constrained strictly to `Yes` or `No`; Mood constrained to emoji list.

10. REPOSITORY ARCHITECTURE & FILE STRUCTURE

```
personal-health-dashboard/
├── README.md                 # Executive summary, portfolio write-up, key insights
├── requirements.md           # Full Product Requirements Document (PRD)
├── test-cases.md             # QA testing framework & edge case test results
├── user-guide.md             # Step-by-step user operational guide
├── google-sheets/
│   └── template-link.txt      # Direct view-only copy link to Google Sheet
└── screenshots/
    ├── dashboard.png          # High-resolution preview of Dashboard tab
    ├── daily-log.png          # High-resolution preview of Daily Log tab
    └── analytics.png          # High-resolution preview of Analytics tab
```

11. DEFINITION OF DONE (DOD CHECKLIST)

Functional Readiness

- [x] Daily input sheet correctly feeds Dashboard.
- [x] 7-day rolling weight average automated.
- [x] Milestone progress triggers dynamically.
- [x] Calorie adherence & averages calculated.

Portfolio & Documentation

- [x] Excel / Google Sheet styled in Blush theme.
- [x] 30-day realistic sample dataset included.
- [x] Test cases executed without formula errors.
- [x] Published to GitHub repository with README.